EE2-2

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2013

EEE PART II: MEng, BEng and ACGI

ANALOGUE ELECTRONICS 2

Wednesday, 12 June 2013 2:00pm

Time allowed: 2 hours

There are THREE questions on this paper.

ALL questions are compulsory.
Question 1 carries 40% of the marks and Questions 2 and 3 carry 30% each.

Any special instructions for invigilators and information for candidates are on page 1.

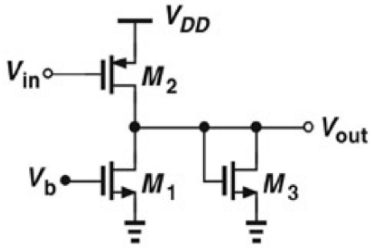
Basic expressions for voltage gain, input and output resistance for single stage amplifiers are given on the last page.

Examiners responsible — First Marker(s): T.G. Constandinou
Second Marker(s): C. Toumazou

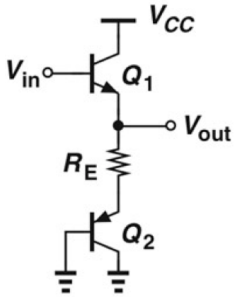
1. This question consists of 6 sub-questions. Each item can be answered in a few words or a short paragraph with use of a diagram and/or expression where helpful. Please answer all sub-questions and simplify expressions where possible stating any assumptions made.

   a) Draw the *small signal model* of a bipolar transistor with parasitic capacitances and explain what limits its high frequency performance when configured as a *common emitter* amplifier. [5]

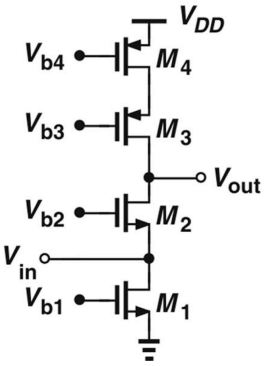
   b) Derive expressions (by inspection) for the *voltage gain* of the amplifier circuits shown below (assuming $\lambda \neq 0$ and $V_A > 0$, i.e. including r_O). [15]



(i)



(ii)

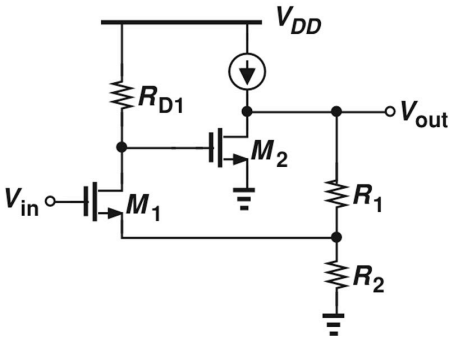


(iii)

   c) Describe what is meant by *common mode rejection ratio (CMRR)* and by using an example application, explain why this is important. [5]

   d) Describe (using a diagram) what is meant by the term *phase margin*. What is a typical phase margin target (in degrees)? [5]

   e) Derive expressions for the *closed loop gain* of the circuit below. Assume $\lambda = 0$. [5]



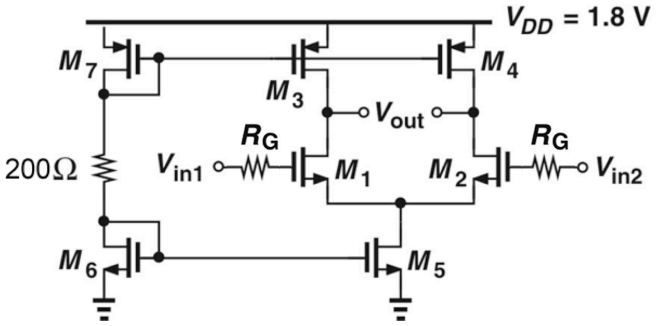
   f) Given an operational amplifier has a *gain-bandwidth product* of 100MHz, an *open-loop gain* of 80dB, maximum *slew rate* of 100kV/s, and rail-to-rail output range, calculate: (i) the -3dB bandwidth, (ii) the full-power bandwidth for a ±3.3V power supply. [5]

2. The circuit shown below is a single-stage fully differential amplifier.

Assume all devices are in saturation and $\lambda>0$ (i.e. $R_{out}<\infty$). Use the following expression for the large signal drain current of a MOSFET (in saturation) with corresponding transistor parameters:

$$I_D = \frac{1}{2}\mu C_{ox}\frac{W}{L}\left(V_{GS}-V_{TH}\right)^2$$

$\mu_n C_{ox} = 200\mu A/V^2$ $\quad V_{THN} = 0.4V$ $\quad \lambda_N = 0.1V^{-1}$
$\mu_p C_{ox} = 100\mu A/V^2$ $\quad V_{THP} = -0.5V$ $\quad \lambda_P = 0.2V^{-1}$



a) Determine expressions for the *differential gain* and *common-mode gain* stating any assumptions made. [6]

b) Calculate $(W/L)_3$, $(W/L)_4$ and $(W/L)_7$ given $I_{D5}=2I_{D6}=2mA$ and $(W/L)_5=2(W/L)_6=100/1$. [4]

c) Calculate $(W/L)_1=(W/L)_2$ for a differential gain of 15. [3]

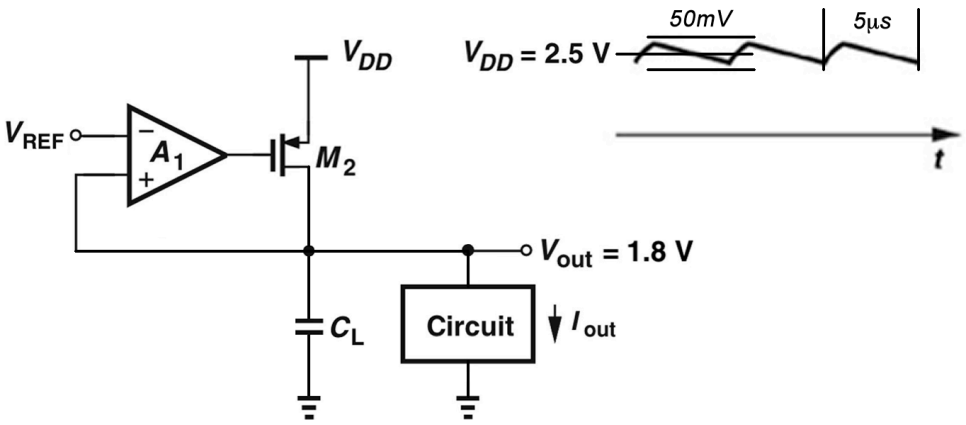
d) Using a half circuit approximation, redraw the differential amplifier circuit above to: (i) identify any nodes that are associated with poles and (ii) include all the parasitic device capacitances. [5]

e) Determine expressions for the pole frequencies stating any assumptions made. Where required, use the following parasitic capacitance values: $C_{GS} = (2/3)WLC_{ox}$, $C_{ox} = 12fF/\mu m^2$, $C_{GD} = C_0W$, $C_0 = 0.2fF/\mu m$, $C_{DB} = C_{SB} = 0$. [5]

f) Given all device lengths ($L_1=L_2=L_3=L_4=L_5=L_6=L_7$) are 1μm and $R_G=2k\Omega$, evaluate the pole frequencies and sketch the bode plot (magnitude response only) for the amplifier. [7]

3. The circuit shown below is a *low dropout (LDO) voltage regulator* for providing a stable DC voltage supply to a load circuit. The required output voltage (V_{out}) is 1.8V with a maximum supply current of 100mA.

   Use the same transistor expressions/parameters given previously in Question 2.



   a) Determine expressions for the *open loop gain* and *feedback factor* of the regulator circuit. [4]

   b) Calculate the low frequency *output impedance* of the voltage regulator, assuming $(W/L)_2$=100/1 and A_1=85dB. [5]

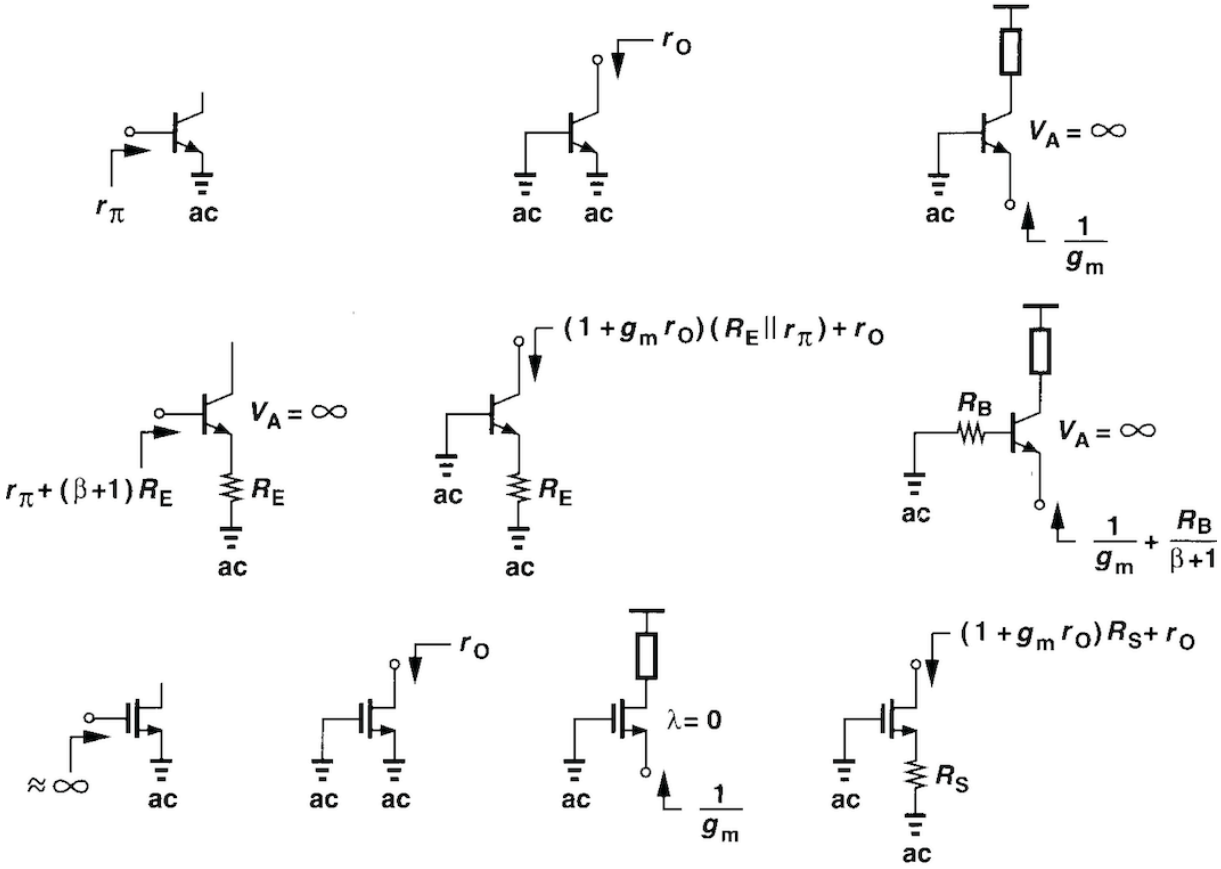
   c) Discuss how the specifications of the operational amplifier need to be selected such that the ripple (shown on right of Figure) is rejected at the regulated output (V_{out}). [5]

   d) Discuss how should the value of C_L be selected? [6]

   e) State three fundamental challenges in designing circuits that generate a reference voltage or current. Describe a reference generator circuit design that overcomes one of these. [6]

   f) A voltage regulator circuit can also be implemented using an NMOS source follower output transistor (for M_2) if the input terminals to the op-amp are swapped (i.e. by connecting V_{REF} to V_+ and feedback to V_-). Give one advantage of each topology. [4]

Input and Output Impedances



Voltage Gain Equations

